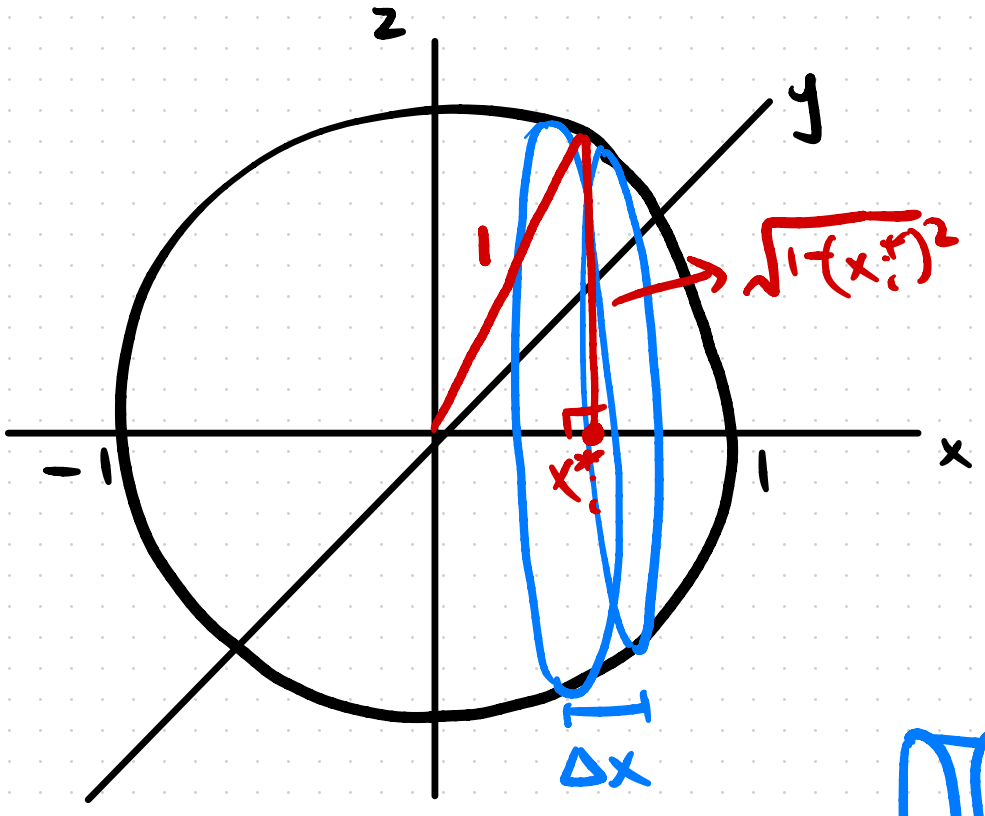
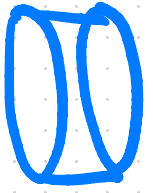
Volumes (6.2)

What is the volume of a sphere with radius 1?



Slice with a plane perpendicular to the x direction.

The solid between the two slices is approximately a cylinder with width Δx, and the radius is $\sqrt{1-(x_i^*)^2}$



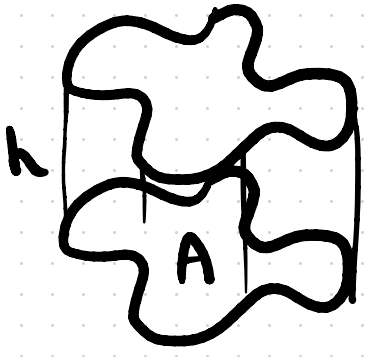
where x_i^* is any sample x-value between the two slices. Therefore the area of the base of the cylinder is

$$\pi\left(\sqrt{1-(x_i^*)^2}\right)^2 = \pi\left(1-(x_i^*)^2\right)$$

The volume of a cylinder / prism is

(area of base) x height

This is true no matter the shape of the base



Volume = Ah

Volume of the cylindrical slice is

$$\pi\left(1-(x_i^*)^2\right)\Delta x$$

Volume of sphere is

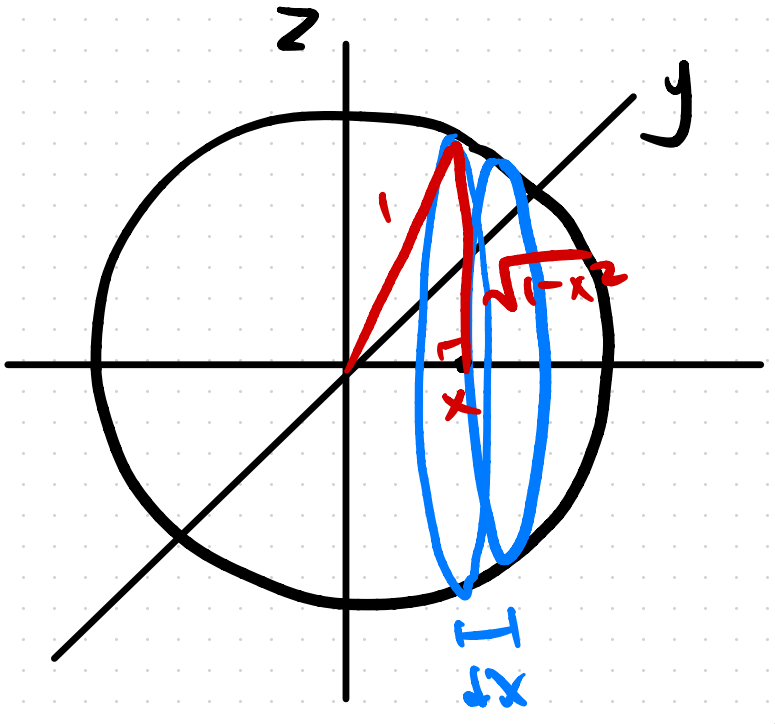
$$\approx \sum_{i=1}^{n} \pi\left(1-(x_i^*)^2\right)\Delta x$$

n = # of cylindrical slices

Volume is exactly

$$\lim_{n \to \infty} \sum_{i=1}^{n} \pi \left(1 - (x_i^*)^2\right) \Delta x$$

$$= \int_{-1}^{1} \pi \left(1 - x^2\right) dx$$



Slice sphere into infinitely thin cylinders with width dx, and the radius of the base is $\sqrt{1-x^2}$

Each "cylinder" has "volume"

$$(\text{base area}) \times \text{width} = \pi\left(\sqrt{1-x^2}\right)^2 dx$$

"Sum" all of these "cylinders", to get

$$\int_{-1}^{1} \pi(1-x^2)\,dx = \pi\left(x-\frac{1}{3}x^3\right)\Big|_{-1}^{1}$$

$$= \frac{2}{3}\pi - \left(-\frac{2}{3}\pi\right)$$

$$= \frac{4}{3}\pi$$

"Cross section" method of finding volumes.

- Pick a direction, say x.
- At each value of x, find the area $A(x)$ of the cross-section at x.
- Evaluate

$$\int_{x\,min}^{x\,max} A(x)\,dx$$

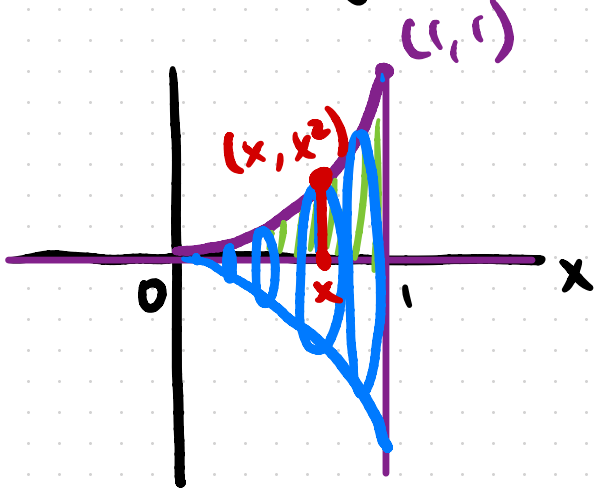
hardest step

"volume" of the cylindrical slice at x

The region bounded by

$x = 1$ $\quad y = 0$ $\quad y = x^2$

is rotated about the x-axis. Find the volume of the resulting solid.



Pick x direction.

The cross-sections are circles.

The cross-section at x is a circle with radius x^2, and area $\pi(x^2)^2 = \pi x^4$.

Volume is

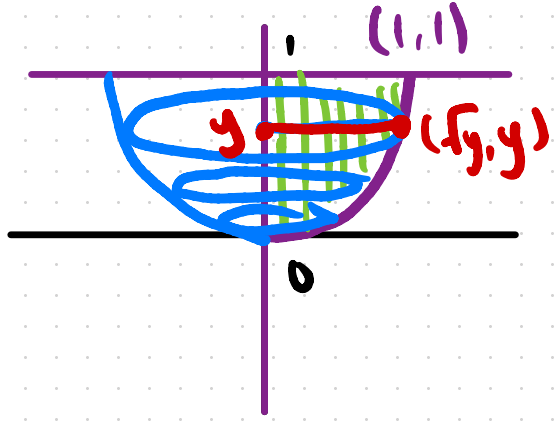
$$\int_0^1 \pi x^4 \, dx = \frac{1}{5}\pi x^5 \Big|_0^1 = \frac{1}{5}\pi$$

The region bounded by

$x = 0$ $\quad y = 1$ $\quad y = x^2$ $\Rightarrow x = \pm\sqrt{y}$

$x = \sqrt{y}$ in 1st quad

is rotated about the y-axis. Find the volume of the resulting solid.



Choose y-direction

Cross-section of y ⊐ a circle with radius $\sqrt{y}$, and area $\pi(\sqrt{y})^2 = \pi y$

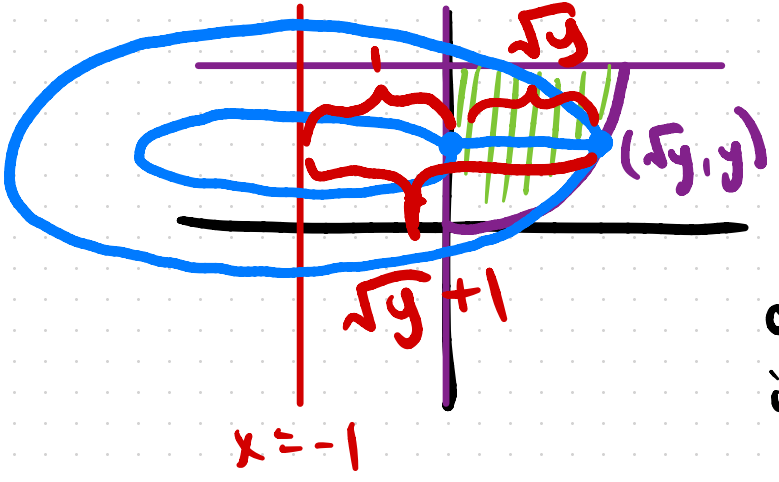
Volume is

$$\int_0^1 \pi y \, dy = \frac{1}{2}\pi y^2 \Big|_0^1 = \frac{1}{2}\pi$$

The region bounded by

$x = 0$ $\quad y = 1$ $\quad y = x^2$

is rotated about the line $x = -1$. Find the volume of the resulting solid.



Pick y-direction.

Cross-sections are annuluses. The cross-section at y has

outer radius = $\sqrt{y} + 1$

inner radius = 1

The area of this cross-section is

$$\left(\begin{array}{c}\text{Area of outer}\\ \text{circle}\end{array}\right) - \left(\begin{array}{c}\text{Area of inner}\\ \text{circle}\end{array}\right)$$

$$\pi\left(\sqrt{y}+1\right)^2 - \pi\left(1\right)^2$$

The volume

$$\int_0^1 \pi\left(\left(\sqrt{y}+1\right)^2 - 1^2\right) dy$$

$$= \pi \int_0^1 \left(y + 2\sqrt{y} \right) dy$$

$$= \pi \left(\frac{1}{2}y^2 + \frac{4}{3}y^{\frac{3}{2}} \right) \Big|_0^1$$

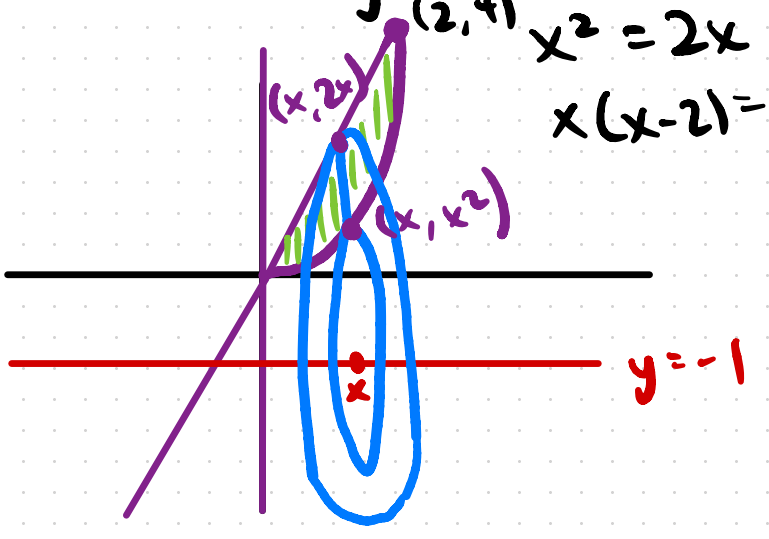
$$= \pi \left(\frac{1}{2} + \frac{4}{3} \right)$$

$$= \frac{11\pi}{6}$$

The region bounded by

$x = 0$ $\quad y = 2x$ $\quad y = x^2$

is rotated about the line $y = -1$. Find the volume of the resulting solid.



$x^2 = 2x$

$x(x-2) = 0$

Pick x-direction.

Cross-sections are annuluses with

outer radius $= 2x+1$

inner radius $= x^2+1$

Area of cross-section is

$$\pi\left((2x+1)^2-(x^2+1)^2\right)$$

Volume

$$\int_0^2 \pi\left((2x+1)^2-(x^2+1)^2\right)dx$$

$$\pi\int_0^2 \left((4x^2+4x+1)-(x^4+2x^2+1)\right)dx$$

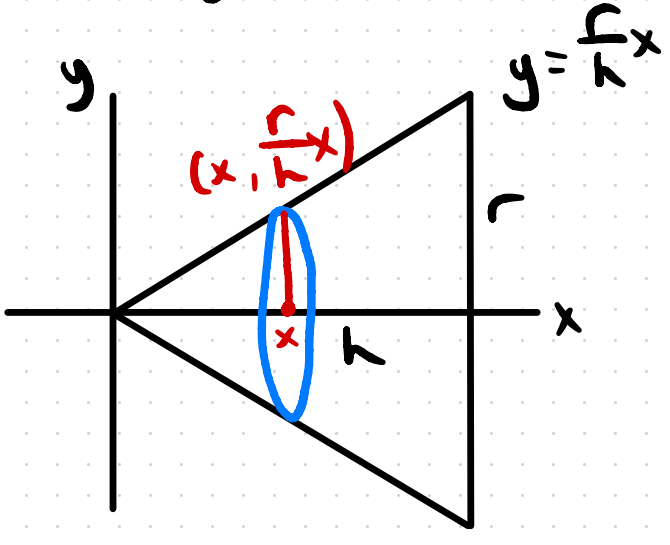
$$\pi \int_0^2 \left(-x^4 + 2x^2 + 4x\right)$$

$$\pi \left(-\frac{1}{5}x^5 + \frac{2}{3}x^3 + 2x^2\right)\Big|_0^2$$

$$\pi \left(-\frac{8}{5} + \frac{16}{3} + 8\right)$$

$$\frac{104\pi}{15}$$

What is the volume of a cone with radius r and height h?



Choose x-direction

Cross-section is circle with radius $\frac{r}{h}x$,

area $\pi\left(\frac{r}{h}x\right)^2$

$$\int_0^h \pi \frac{r^2}{h^2} x^2 \, dx = \frac{1}{3}\pi \frac{r^2}{h^2} x^3 \Big|_0^h = \frac{1}{3}\pi r^2 h$$